Shared-Private Decoupling-Based Multilevel Feature Alignment Semisupervised Learning for HSI and LiDAR Classification

Jiahui Qu, *Member, IEEE*, Lijian Zhang, Wenqian Dong, *Member, IEEE*, Nan Li, and Yunsong Li, *Member, IEEE*

***Abstract*— The joint classification methods of hyperspectral image (HSI) and light detection and ranging (LiDAR) data based on deep learning have demonstrated exceptional classification performance with sufficient labeled samples. However, it is expensive and time-consuming to acquire labeled data. To address this limitation, we propose a shared-private decoupling-based multilevel feature alignment semisupervised (SASS) learning method for HSI and LiDAR classification, which introduces the idea of domain adaptation (DA) to capture the shared features of labeled and unlabeled data for classification and circularly selects reliable pseudolabels based on these features to retrain the model. Specifically, we treat labeled data as the source domain (SD) and unlabeled data as the target domain (TD) and propose a shared-private feature decoupling (SPFD) module to acquire shared representations of SD and TD by separating domain private features. The multilevel shared feature alignment (MSFA) strategy is designed to synthetically consider both spatial details and semantic information by minimizing the maximum mean discrepancy (MMD) between these shared features. In addition, we design a graph transformer-based class-balanced pseudolabel generation (GBPG) strategy for iterative model training with reliable pseudolabels, which exploits the graph transformer network-based sample acquisition (GTSA) strategy to select valuable samples and generate corresponding pseudolabels using the adaptive class-specific threshold-based sample annotation (ATSA) strategy. Experimental results on three public datasets validate the effectiveness of the proposed method. The code is available at https://github.com/Jiahuiqu/SASS.**

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***Index Terms*— Active learning, domain adaptation (DA), hyperspectral image (HSI), image classification, light detection and ranging (LiDAR) data, semisupervised.**

I. Introduction

REMOTE sensing (RS) images have extensive applications across various fields, including environmental protection, urban planning, and resource exploration [1], [2], [3]. In various RS images, hyperspectral images (HSIs) can capture the response of ground objects in different wavelength bands. Therefore, HSI has significant advantages in distinguishing ground objects of various materials and has become one of the most important data sources in RS tasks [4], [5], [6], [7]. However, it performs poorly in distinguishing land cover types with the same spectral characteristics but different heights. Light detection and ranging (LiDAR) is an active mapping technology that provides high-precision ground elevation information for distinguishing ground objects at various heights [8]. However, for objects with the same height but different materials, its discriminative ability may be limited. Therefore, the joint classification of multisource data can leverage their advantages to further improve classification accuracy [9], [10].

Inspired by the success of convolutional neural networks (CNNs) in the field of computer vision, numerous CNN-based methods for joint classification have been proposed. Chen et al. [11] proposed a feature fusion framework based on CNN to capture multisource data features, which explores a new path for multisource data classification. Hong et al. [12] presented a framework that follows a deep encoder–decoder network architecture (EndNet), which makes full use of the advantages of the encoder–decoder module to fuse features from multisource data. To tackle the semantic gap problem when integrating modalities, Xia et al. [13] designed a multimodal contrastive learning framework, which achieves semantic feature alignment across modalities. Zhang et al. [14] proposed a novel multimodal adaptive modulation network (MAMNet) to make full use of the complementary characteristics of different modalities. Jia et al. [15] designed a fusion method based on collaborative contrastive learning (CCL), which achieves coordinated feature representation from multisource data without labeled samples. For effectively exploiting

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the information of multisource data, Li et al. [16] proposed A^3CLNN, which contains a three-level fusion strategy to fully integrate multisource data features. Feng et al. [17] proposed a novel joint classification method, which combines pixel-level label assignment and decision-level label prediction to achieve high-reliability land cover classification. Although CNN-based methods are good at learning local spatial features, they may not effectively capture global spectral information.

Recently, transformers have been introduced to the field of RS classification due to the unique advantages of exploiting global features and achieved excellent results [18], [19]. Ding et al. [20] designed a novel global–local transformer network (GLT-Net) to extract local and global spatial–spectral relationships, establishing an effective interaction between HSI and LiDAR data. Xue et al. [21] proposed a deep hierarchical vision transformer (DHViT) architecture, which fuses heterogeneous features from multisource data with the cross-attention feature fusion module to improve the performance. Roy et al. [22] designed a multihead cross-patch attention mechanism for RS data fusion. Zhao et al. [23] proposed a dual-branch method, which fuses different features effectively using a cross-token attention fusion encoder. Yang et al. [24] proposed a text-supervised multidimensional contrastive fusion network (TMCFN) to capture rich semantic information in complex data distributions, which leverages class text information to guide the learning of vision features and establishes semantic connections between them for classification. To address the problem of heterogeneity between multimodal data, Hu et al. [25] designed a global clue-guided cross-memory quaternion transformer network (GCCQTNet), which leverages global clue to guide multimodal feature learning and facilitate cross-modality interaction. Yang et al. [26] proposed a modality fusion vision transformer (MFViT), which integrates heterogeneous features from HSI and LiDAR data, enabling comprehensive utilization of multimodal information.

Generally, the methods mentioned above are usually supervised algorithms. They can achieve superior performance with ample training data. However, their performance will significantly degrade when training data are limited. As we all know, pixel-level labeling in the field of RS is expensive and time-consuming. These difficulties hinder the application of supervised methods. Therefore, it is an important research topic to achieve high-precision classification with limited labeled samples. In this article, we propose a shared-private decoupling-based multilevel feature alignment semisupervised (SASS) learning method for HSI and LiDAR classification with limited labeled samples. The main contributions can be summarized as follows.

1) We design SASS for HSI and LiDAR classification with limited labeled samples, which introduces the idea of DA to align the feature distribution of labeled and unlabeled data and iteratively selects valuable pseudolabels to improve classification performance.
2) Inspired by the idea of DA, we treat labeled data as the source domain (SD) and unlabeled data as the target domain (TD), respectively. We design a shared-private feature decoupling (SPFD) module to capture the shared features across different domains. The multilevel shared feature alignment (MSFA) strategy is proposed to align both semantic information and spatial details to make the classifier trained on shared features better fit the distribution of the TD.
3) To avoid the overfitting phenomenon caused by pseudolabels providing similar information as labeled samples, we propose a graph transformer-based class-balanced pseudolabel generation (GBPG) strategy, which leverages the graph transformer network-based sample acquisition (GTSA) strategy to select valuable unlabeled samples and generates corresponding pseudolabels using the adaptive class-specific threshold-based sample annotation (ATSA) strategy.

The remaining parts can be organized as follows. Section II provides a brief description of domain adaptation (DA) and semisupervised learning (SSL). Section III introduces the details of SASS. Section IV illustrates the results of experiments and ablation analysis. Finally, the conclusion is summarized in Section V.

II. Related Work

A. Domain Adaptation

A direct approach to solve the lack of labeled data in the TD is to transfer a model train on the SD to the TD [27]. However, the model performs poorly when directly applied to the TD due to the distribution discrepancy between different domains. Most existing DA methods achieve domain alignment through techniques such as metric learning and domain adversarial training. The methods based on metric learning focus on learning invariant representation by minimizing the difference metric of the feature distribution. The maximum mean discrepancy (MMD), correlation alignment (CORAL), and graph optimal transport (GOT) have been introduced to align feature distributions. Zhang et al. [28] designed a multilevel DA network (MDA-Net), which utilizes optimal transport (OT) and heterogeneous knowledge graph (HKG) matching to achieve distribution alignment. Dong et al. [29] proposed a contrastive constrained cross-scene model-informed interpretable classification strategy (C3MI-C) to align the feature distribution across different domains. Adversarial training-based methods aim to make features from different domains indistinguishable regarding domain labels, which is achieved through an adversarial loss. Wang et al. [30] designed a biclassifier adversarial network (BCAN), which combines confrontational learning and collaborative learning to facilitate knowledge transfer from SD to TD. Huang et al. [31] proposed a two-branch attention adversarial DA (TAADA) network to enhance cross-domain classification based on adversarial learning. The authors then proposed a spatial–spectral weighted adversarial DA (SSWADA) network [32] and a framework based on calibrated prototype and dynamic instance convolution (CPDIC) [33], both employing adversarial learning strategies. Inspired by the idea of DA, we consider labeled data as the SD and unlabeled data as the TD, aiming to capture their shared feature representations to improve the ability of the classifier to fit the distribution of the TD.

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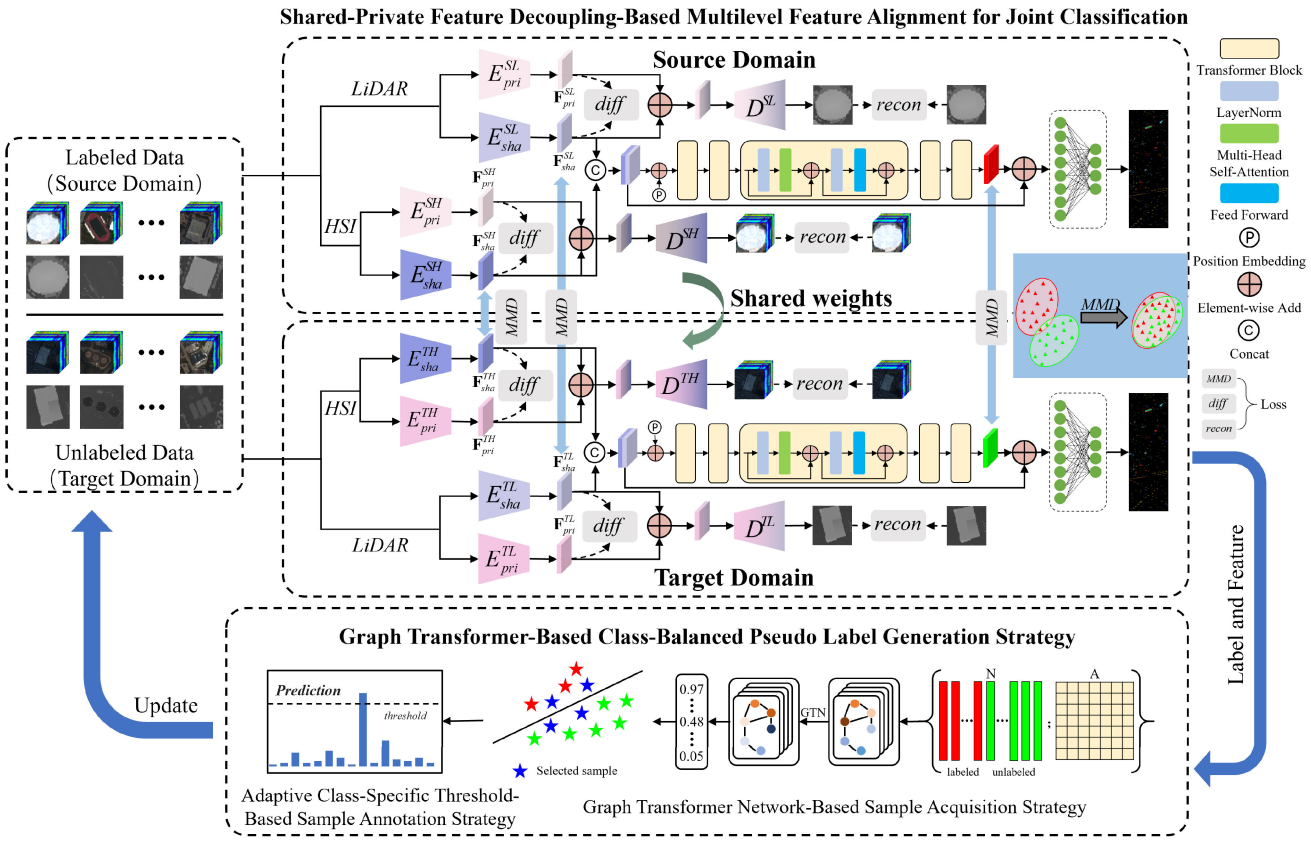


Fig. 1. Architecture of the proposed SASS learning method for HSI and LiDAR classification. SASS employs the idea of DA to learn shared feature representations to enhance the ability of the model to fit the distribution of unlabeled data. Based on these features, the graph transformer network-based class-balanced pseudolabel generation strategy is designed to select valuable pseudolabels for model retraining, ensuring effective learning from unlabeled data.

B. Semisupervised Learning

To improve the performance of the model trained with limited labeled samples, SSL not only captures features from labeled data but also leverages the information from unlabeled data. Entropy minimization and consistency regularization are the most common strategies in SSL. The entropy minimization encourages the model to produce predictions with low entropy. Some methods regard predicted probability as confidence and select high-confidence pseudolabels to retrain the model. Li et al. [34] designed a triplet semisupervised deep network (TSDN), which generates high-confidence pseudolabels by comparing multiview classification results. Lu et al. [35] proposed a dual-stream class-adaptive network (DSCA-Net), which adaptively determines thresholds for different classes based on their classification difficulties. Chen et al. [36] proposed an adaptive pseudolabel feature learning (APFL) model, which utilizes iterative multiscale superpixel segmentation to generate pseudolabels. Consistency regularization assumes that the model predictions should remain relatively stable under perturbations. Liu et al. [37] proposed contrastive mutual learning with pseudolabel smoothing (CMLP), which avoids the overlearning of noise data during training. Zhao et al. [38] designed a semisupervised cotraining model using convolution and transformer (SCM-CT), which learns valuable spectral–spatial information from conflict predictions to better utilize unlabeled samples.

III. Methods

As shown in Fig. 1, SASS is designed for HSI and LiDAR classification, which acquires domain-shared features for classification and employs self-training with selected pseudolabels. We decouple the input patch into the domain-shared and private feature spaces. The shared features are utilized to train the classifier. In addition, we design a GTSA strategy to iteratively select valuable samples and generate corresponding pseudolabels using the adaptive class-specific threshold. Finally, we leverage high-confidence pseudolabels to update the labeled data repeatedly and conduct self-training to improve the classification performance.

A. Shared-Private Decoupling-Based Multilevel Feature Alignment (SPFA) for Joint Classification

Because of the different distribution of the labeled and unlabeled data, the model trained with limited labeled samples typically performs poorly on unlabeled data. DA methods, which minimize the distribution discrepancy between the SD and the TD to improve the classification performance of TD, have also significantly advanced in recent years. Inspired by the idea of DA, we treat labeled data as SD and unlabeled data as TD and align the feature distributions from different domains to make the model perform better on TD. Previous research focuses on discovering representations that are shared

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between two domains. However, this approach leaves the shared features vulnerable to contamination by noise that correlates with the underlying shared distribution. Therefore, we model shared and private components of the domain representations and use the shared features to train the classifier to make it better fit the distribution of TD. The process is divided into three parts: SPFD, vision transformer-based shared feature refinement (ViTFR), and MSFA.

1) Shared-Private Feature Decoupling: To enhance domain-shared representation learning for joint classification, we design SPFD, which generates a shared representation for both domains while also producing distinct private representations for each domain. By partitioning the space in this manner, the classifier trained on shared features can better generalize across domains as its inputs are uncontaminated by domain private features. The SPFD consists of three blocks: the shared encoder, the private encoder, and the decoder. The shared and private encoders are designed to capture domain-shared and private features, respectively. We align the shared features across different domains and apply constraints on the shared and private features within the same domain to promote their independence, thereby achieving effective feature decoupling. The decoder is designed to reconstruct the original data using both private and shared features, ensuring that the extracted features retain valuable information from the original data.

To capture the spatial–spectral features of HSI and the spatial-elevation information of LiDAR, we design different feature extractors for each modality. Specifically, the numerous channels in the HSI modality result in deep networks having a significant number of parameters. While training with limited labeled samples, they often suffer from overfitting problems, leading to poor generalization on unlabeled data. The residual neural network (ResNet) is known for its robust spatial–spectral feature learning capabilities due to the incorporation of skip connections. Therefore, we apply the ResNet structure to learn domain-shared and private spectral–spatial features from the input HSI patches. The ResNet contains an AvgPool layer and two ResBlocks, which consist of two Conv-BN-LeakyReLU blocks and a skip connection. For LiDAR, we use CascadeNet [39] to extract domain-shared and private spatial-elevation features from the input LiDAR patches. The CascadeNet contains a Conv-BN-LeakyReLU block, an AvgPool layer, and a cascade block, which consists of a Conv-Conv block, two LeakyReLU layers, a Conv-BN-Conv block, and two skip connections. The cascade operation combines features from different layers to facilitate feature reuse and propagation. The process of extracting domain-shared features $\{\mathbf{F}_{\text{sha}}^{\text{SH}(L)}, \mathbf{F}_{\text{sha}}^{\text{TH}(L)}\}$ and domain private features $\{\mathbf{F}_{\text{pri}}^{\text{SH}(L)}, \mathbf{F}_{\text{pri}}^{\text{TH}(L)}\}$ can be expressed as follows:

$$\begin{cases} \mathbf{F}_{\text{pri}}^{\text{SH}(L)} = E_{\text{pri}}^{\text{SH}(L)}\left(\mathbf{X}^{\text{SH}(L)}\right) \\ \mathbf{F}_{\text{sha}}^{\text{SH}(L)} = E_{\text{sha}}^{\text{SH}(L)}\left(\mathbf{X}^{\text{SH}(L)}\right) \\ \mathbf{F}_{\text{sha}}^{\text{TH}(L)} = E_{\text{sha}}^{\text{TH}(L)}\left(\mathbf{X}^{\text{TH}(L)}\right) \\ \mathbf{F}_{\text{pri}}^{\text{TH}(L)} = E_{\text{pri}}^{\text{TH}(L)}\left(\mathbf{X}^{\text{TH}(L)}\right) \end{cases} \tag{1}$$

Algorithm 1 Proposed SASS

Input: labeled dataset D_s; unlabeled dataset D_t; AL stages R; decision boundary d;

```
for iter = 1 to R do
    Train the classifier using shared features of D_s and D_t.
    Save the prediction probabilities p.
    Initialize the queried set D_q and the pseudo label set D_p as empty.
    Initialize nodes V with the shared features
    Calculate the adjacency matrix A by Eq. (12).
    Train the GTN using V and A.
    for i = 1 to n do
        D_q = D_q ∪ {argmin_{x_u} |d − q(x_u)|}
    end
    Calculate the global threshold th_g by Eq. (18).
    Calculate the class-specific threshold th_c by Eq. (20).
    D_p = D_p ∪ {x_i | p_i(c) > th_c ∧ x_i ∈ D_q}
    D_s = D_s ∪ D_p
    D_t = D_t − D_p
end
```

Output: the classification result.

where $\mathbf{X}^{\text{SH}}$ and $\mathbf{X}^{\text{TH}}$ represent HSI patches from SD and TD, $\mathbf{X}^{\text{SL}}$ and $\mathbf{X}^{\text{TL}}$ represent LiDAR patches from SD and TD, $E_{\text{sha}}^{\text{SH}(L)}$ and $E_{\text{sha}}^{\text{TH}(L)}$ represent the domain-shared feature encoders, and $E_{\text{pri}}^{\text{SH}(L)}$ and $E_{\text{pri}}^{\text{TH}(L)}$ represent domain private feature encoders.

$E_{\text{sha}}^{\text{SH}(L)}(\cdot)$ and $E_{\text{sha}}^{\text{TH}(L)}(\cdot)$ are consistent with each other through weight sharing, which significantly reduces the number of parameters. $E_{\text{pri}}^{\text{SH}(L)}(\cdot)$ and $E_{\text{pri}}^{\text{TH}(L)}(\cdot)$ are utilized to obtain the domain-specific features, which have the same structure as the domain-shared encoder but with different weights. The shared decoder $D(\cdot)$ is designed to reconstruct the origin data using both shared and private features, ensuring that the extracted features preserve crucial information from input data. The process is given as follows:

$$\begin{cases} \widehat{\mathbf{X}}^{\text{SH}(L)} = D\left(\mathbf{F}_{\text{sha}}^{\text{SH}(L)} + \mathbf{F}_{\text{pri}}^{\text{SH}(L)}\right) \\ \widehat{\mathbf{X}}^{\text{TH}(L)} = D\left(\mathbf{F}_{\text{sha}}^{\text{TH}(L)} + \mathbf{F}_{\text{pri}}^{\text{TH}(L)}\right) \end{cases} \tag{2}$$

where $\widehat{\mathbf{X}}^{\text{SH}(L)}$ and $\widehat{\mathbf{X}}^{\text{TH}(L)}$ represent the reconstructed samples of SD and TD.

The fusion features can be obtained by concatenating the shared spatial–spectral features of HSI and the shared spatial-elevation features of LiDAR. The process is given as follows:

$$\mathbf{F}_{\text{fus}}^{S(T)} = \text{Concat}\left(\mathbf{F}_{\text{sha}}^{S(T)H}, \mathbf{F}_{\text{sha}}^{S(T)L}\right) \tag{3}$$

where $\text{Concat}(\cdot)$ represents the concatenation operation and $\mathbf{F}_{\text{fus}}^{S}$ and $\mathbf{F}_{\text{fus}}^{T}$ denote the fusion features of SD and TD, respectively.

2) Vision Transformer-Based Shared Feature Refinement: Although CNN excels at capturing local spatial features, it may struggle to leverage global information effectively. The vision transformer (ViT) has powerful global modeling capabilities,

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enabling it to model deep semantic relationships to complement local information. Taking a series of tokens as input, ViT employs the multihead self-attention (MHSA) mechanism to capture intertoken relationships. Therefore, we design the ViT-based shared feature refinement (ViTFR) module to enhance semantic information in the fusion features.

The structure of ViTFR is shown in Fig. 1. Initially, we superimpose the position embedding into $\mathbf{F}_{\text{fus}}^{S(T)}$ to acquire the input $\mathbf{F}_0^{S(T)}$ for ViT. The process of going through the ViT is given as follows:

$$\begin{aligned}\widehat{\mathbf{F}}_l^{S(T)} &= \text{MHSA}\left(\text{LN}\left(\mathbf{F}_{l-1}^{S(T)}\right)\right) + \mathbf{F}_{l-1}^{S(T)} \\ \mathbf{F}_l^{S(T)} &= \text{FFN}\left(\text{LN}\left(\widehat{\mathbf{F}}_l^{S(T)}\right)\right) + \widehat{\mathbf{F}}_l^{S(T)}\end{aligned} \tag{4}$$

where $l = 1, 2, \ldots, L$; L signifies the depth of ViT; MHSA$(\cdot)$, LN$(\cdot)$, and FFN$(\cdot)$ represent the MHSA, layer normalization operation, and feedforward network, respectively; and $\mathbf{F}_l^{S(T)}$ represents the output features of the lth transformer block. The output of ViT is added to $\mathbf{F}_{\text{fus}}^{S(T)}$ to supplement the semantic information in the fusion features.

We employ the MHSA mechanism to enhance the semantic information. Self-attention (SA) is the core of MHSA. Initially, the feature tokens undergo linear combinations with three learnable weights $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, resulting in three distinct matrices named query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$). Subsequently, we use matrix $\mathbf{Q}$ and $\mathbf{K}$ to obtain the attention score, which is then transformed into weight probabilities via the softmax function. Finally, self-attention is achieved via multiplying the score with $\mathbf{V}$. The process is defined as

$$\text{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d}}\right)\mathbf{V} \tag{5}$$

where d represents the number of channels in input tokens. The MHSA can efficiently aggregate information from various SA modules to capture more comprehensive features. The process of MHSA is illustrated as follows:

$$\begin{aligned}\text{MHSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{Concat}(a_1, \ldots, a_j, \ldots, a_s)\mathbf{W}_O \\ \text{where } a_j &= SA(\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j)\end{aligned} \tag{6}$$

where s represents the number of attention heads and $\mathbf{W}_O$ is the parameter matrix.

3) Multilevel Shared Feature Alignment: Most DA methods focus on aligning high-level features. Although these approaches realize domain transfer learning to a certain extent, they ignore low-level features with spatial information. Therefore, we design the MSFA to alleviate this problem.

We reduce the distribution discrepancy by minimizing the MMD between shared fusion features, encouraging them to be as similar as possible. The MMD loss is defined as

$$L_{\text{MMD}} = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi\left(\mathbf{F}_{\text{fus,i}}^{S}\right) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi\left(\mathbf{F}_{\text{fus,j}}^{T}\right)\right\|_{\mathcal{H}}^2 \tag{7}$$

where $\phi(\cdot)$ represents a mapping function from the original feature space to the reproducing kernel Hilbert space (RKHS), $\|\cdot\|_{\mathcal{H}}$ represents the RKHS norm, and $\mathbf{F}_{\text{fus,i}}^{S}$ and $\mathbf{F}_{\text{fus,j}}^{T}$ represent the features of each sample. In addition, we use the same way to align the spatial–spectral features, the spatial-elevation features, and the semantic information of the SD and the TD.

Compared with methods that only align high-level feature distributions, the proposed MSFA considers the differences in feature distributions at each level and ensures efficient learning of semantic information and spatial details.

4) Objective Function: The optimization objectives of the proposed classification model are listed as follows.

1) *Classification Loss:* The fusion features containing abundant spatial and semantic information are fed into a fully connected layer to predict the probability distribution of labeled samples. The classification loss is calculated as follows:

$$L_{\text{cls}} = -\frac{1}{n_s}\sum_{i=1}^{n_s}\boldsymbol{y}_i \log \boldsymbol{p}_i \tag{8}$$

where $\boldsymbol{y}_i$ is the one-hot encoding of the class label and $\boldsymbol{p}_i$ is the prediction result.

2) *Difference Loss:* We use a soft subspace orthogonality constraint [40] between the shared and private features of each domain to encourage the independence of these features. The loss L_{diff} is defined as

$$L_{\text{diff}} = \left\|\left(\mathbf{F}_{\text{sha}}^{\text{SH}(L)}\right)^{\text{T}}\mathbf{F}_{\text{pri}}^{\text{SH}(L)}\right\|_F^2 + \left\|\left(\mathbf{F}_{\text{sha}}^{\text{TH}(L)}\right)^{\text{T}}\mathbf{F}_{\text{pri}}^{\text{TH}(L)}\right\|_F^2 \tag{9}$$

where $\|\cdot\|_F$ represents the squared Frobenius norm and T is the transpose operation.

3) *Reconstruction Loss:* We employ the scale-invariant mean squared error between the original sample and the corresponding reconstructed sample [41], which enables the model to concentrate on reproducing the overall shape of the input patch without expending resources on the intensity. The reconstruction loss L_{recon} can be expressed as

$$\begin{aligned}L_{\text{recon}} = &\frac{1}{n_s k}\sum_{i=1}^{n_s}\left\|\mathbf{X}_i^{\text{SH}(L)} - \widehat{\mathbf{X}}_i^{\text{SH}(L)}\right\|_2^2 \\ &- \frac{1}{n_s k^2}\sum_{i=1}^{n_s}\left(\sum\left(\mathbf{X}_i^{\text{SH}(L)} - \widehat{\mathbf{X}}_i^{\text{SH}(L)}\right)\right)^2 \\ &+ \frac{1}{n_t k}\sum_{j=1}^{n_t}\left\|\mathbf{X}_j^{\text{TH}(L)} - \widehat{\mathbf{X}}_j^{\text{TH}(L)}\right\|_2^2 \\ &- \frac{1}{n_t k^2}\sum_{j=1}^{n_t}\left(\sum\left(\mathbf{X}_j^{\text{TH}(L)} - \widehat{\mathbf{X}}_j^{\text{TH}(L)}\right)\right)^2\end{aligned} \tag{10}$$

where n_s and n_t are the number of samples for SD and TD, $\mathbf{X}_i^{\text{SH}(L)}$ and $\mathbf{X}_j^{\text{TH}(L)}$ represent the ith sample in SD and the jth sample in TD, $\widehat{\mathbf{X}}_i^{\text{SH}(L)}$ and $\widehat{\mathbf{X}}_j^{\text{TH}(L)}$ represent the corresponding reconstructed sample, $\sum$ represents the summation of all elements in the matrix, and k is the number of elements in the input patch.

4) The total objective function can be calculated as

$$L = L_{\text{cls}} + \alpha L_{\text{mmd}} + \beta L_{\text{diff}} + \gamma L_{\text{recon}} \tag{11}$$

where α, β, and γ are weight coefficients.

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B. Graph Transformer-Based Class-Balanced Pseudolabel Generation Strategy

For HSI and LiDAR classification, manual pixel-level annotation is costly and labor-intensive. An effective semisupervised approach to mitigate this issue involves selecting high-confidence pseudolabels for model retraining. However, these pseudolabels may contain information similar to the labeled data, thus leading to potential overfitting. To address this problem, we design GBPG, which leverages the graph transformer network (GTN) to distinguish labeled and unlabeled samples based on the shared fusion features. According to the output of GTN, we select unlabeled samples that cannot be distinguished by GTN to generate a pseudolabel using the class-specific threshold.

1) GTSA Strategy: We design GTSA to select valuable samples. First, we construct the graph with the shared fusion features of SD and TD. Then, the GTN is employed to learn the relationship between different nodes and to distinguish labeled and unlabeled samples. According to the output of GTN, the unlabeled samples situated near the decision boundary are selected to generate corresponding pseudolabels.

The graph consists of nodes $\mathbf{V}$ and the edges. The nodes are initialized using the shared features of corresponding samples, and the edges reflect the relationships between graph nodes, typically represented with an adjacency matrix $\mathbf{A}$, which can be defined as

$$\mathbf{A}_{ij} = \begin{cases} 1, & j \in N_i \text{ or } i \in N_j \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where N_i and N_j denote the sets of neighbors of nodes i and j, respectively. We select 20 graph nodes that exhibit the highest similarity to the central node as adjacent nodes. After performing L_2 normalization on $\mathbf{V}$, the similarity between nodes can be defined as

$$\text{sim}(i, j) = 2 - 2\frac{\langle z_i, z_j \rangle}{\|z_i\|_2 \|z_j\|_2} \tag{13}$$

where z_i and z_j, respectively, represent the feature of nodes i and j.

The GTN is utilized to learn the relationships between nodes in the graph. As shown in Fig. 2, the node update equations of the lth layer in GTN can be expressed as

$$\begin{aligned} \widehat{z}_i^{l+1} &= \text{Norm}\left(z_i^l + \widehat{z}_i^l\right) \\ z_i^{l+1} &= \text{Norm}\left(\mathbf{W}_2^l \text{ReLU}\left(\mathbf{W}_1^l \widehat{z}_i^{l+1}\right)\right) \end{aligned} \tag{14}$$

where z_i^l represents the feature of node i at the lth layer, $l = 0, 1$, $\mathbf{W}_1^l$ and $\mathbf{W}_2^l$ are learned parameter matrices, Norm is the normalization operation, ReLU is the activation function, and $\widehat{z}_i^l$ is defined as

$$\begin{aligned} \widehat{z}_i^l &= \mathbf{W}_O^l \text{Concat}\left(\sum_{j \in N_i} a_{ij}^{h,l}\right) \\ \text{where } a_{ij}^{h,l} &= \text{softmax}\left(\frac{\boldsymbol{q}_j^{h,l}(\boldsymbol{k}_i^{h,l})^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{v}_i^{h,l} \\ &= \text{softmax}\left(\frac{\mathbf{W}_Q^{h,l} z_j^l (\mathbf{W}_K^{h,l} z_i^l)^{\mathrm{T}}}{\sqrt{d}}\right)\mathbf{W}_V^{h,l} z_i^l \end{aligned} \tag{15}$$

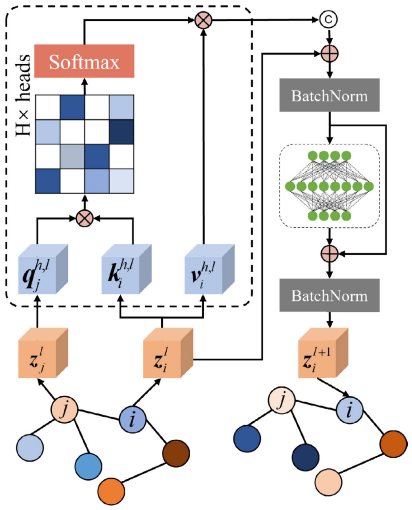


Fig. 2. Structure of the graph transformer layer.

where $\mathbf{W}_O^l$, $\mathbf{W}_Q^{h,l}$, $\mathbf{W}_K^{h,l}$, and $\mathbf{W}_V^{h,l}$ are learned weight matrices at the lth layer; $\boldsymbol{q}_j^{h,l}$, $\boldsymbol{k}_i^{h,l}$, and $\boldsymbol{v}_i^{h,l}$, respectively, represent the query, key, and value vectors; $h = 1, 2, \ldots, s$; and s is the number of attention heads.

The node representations of the final layer will be fed into the fully connected layer to obtain prediction probabilities $\boldsymbol{q}$ of length N, with values ranging between 0 and 1. The probability close to 1 indicates that the sample is similar to the labeled ones. We optimize this objective using binary cross-entropy (BCE) loss [42]

$$L_{\text{BCE}} = -\frac{1}{N_l}\sum_{i=1}^{N_l}\log(\boldsymbol{q}(x_i)) - \frac{\lambda}{N_u}\sum_{j=1}^{N_u}\log(1 - \boldsymbol{q}(x_j)) \tag{16}$$

where λ is the weight factor between two terms, and N_l and N_u, respectively, represent the number of labeled and unlabeled samples in the kth stage.

Given the output $\boldsymbol{q}$ of GTN, we can select valuable unlabeled samples. Suppose that $\boldsymbol{q}(x_u)$ denotes the confidence of unlabeled sample x_u. When $\boldsymbol{q}(x_u)$ approaches 1, it indicates that the sample is easy to distinguish and has similar information as labeled ones. Conversely, when $\boldsymbol{q}(x_u)$ approaches 0, it suggests that the sample is challenging for the classifier to discriminate and may be beneficial for improving the classification performance. Thus, we select unlabeled samples that are situated near the decision boundary d because they are highly uncertain and easy to identify. The sampling process can be described as

$$D_q = D_q \cup \left\{\arg\min_{x_u} |d - \boldsymbol{q}(x_u)|\right\} \tag{17}$$

where D_q represents the queried set.

2) ATSA Strategy: After acquiring the queried set D_q, we select high-confidence pseudolabels to retrain the model. However, using a fixed threshold may lead the network to prioritize easily distinguishable classes, neglecting those more challenging ones. Consequently, the model may encounter overfitting issues for easy categories and bias predictions for hard categories [35]. Therefore, we introduce ATSA to

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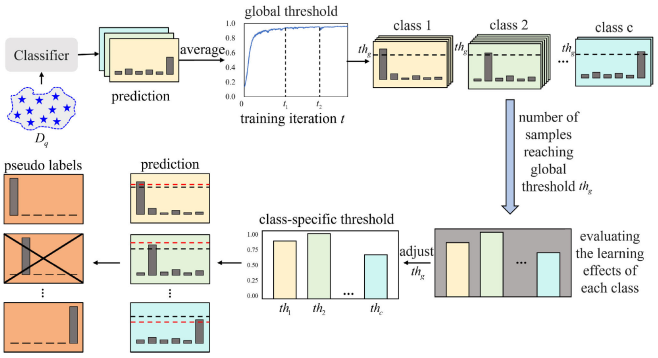


Fig. 3. Process of the ATSA strategy. The global threshold is established based on the prediction results of unlabeled samples in the queried set to effectively reflect the overall learning status. This threshold is then dynamically adjusted according to the classification difficulty of different classes to mitigate overfitting in classes with abundant samples and biased predictions in classes with limited samples, keeping the sample number of each class in a balanced state.

estimate distinct thresholds for different classes. Initially, we establish a global confidence threshold with the prediction results of unlabeled samples in D_q. Subsequently, we define class-specific thresholds based on the count of samples per class whose predictions meet the global threshold. The process is illustrated in Fig. 3.

The global threshold should effectively reflect the overall learning status. Therefore, we set the average of prediction results from the unlabeled samples in D_q as the global threshold th_g. The global threshold th_g can be expressed as

$$\text{th}_g = \frac{1}{|D_q|} \sum_{i=1}^{|D_q|} \max(\boldsymbol{p}_i) \tag{18}$$

where $|D_q|$ represents the number of samples in D_q and $\boldsymbol{p}_i$ denotes the prediction results of unlabeled samples.

Since classification difficulty varies for different classes, it is necessary to design thresholds for each class. A high threshold can effectively filter out noisy pseudolabels, retaining only high-quality labels. Therefore, the count of samples that are predicted to belong to a class and reach the threshold can indicate the learning effect for that class. Consequently, classes with fewer samples surpassing the threshold are deemed more challenging to learn. The number $\text{num}(c)$ of samples belonging to class c is defined as

$$\text{num}(c) = \sum_{i}^{|D_q|} I(\max(\boldsymbol{p}_i) > \text{th}_g) I(\arg\max(\boldsymbol{p}_i) = c) \tag{19}$$

where $I(\cdot)$ signifies that the output result is 1 when the condition is met, and num is normalized by $\lambda_c = \text{num}(c)/\overline{\text{num}}$, where $\overline{\text{num}}$ denotes the average of num. It is evident that classes with fewer samples have lower λ_c values. We use λ_c to adjust the global threshold th_g as follows:

$$\text{th}_c = \begin{cases} 0.95, & \text{if } \lambda_c \times \text{th}_g > 0.95 \\ \lambda_c \times \text{th}_g, & \text{otherwise} \end{cases} \tag{20}$$

where th_c represents the threshold for class c. To guarantee that each class has an adequate number of samples, we adjust the threshold according to the count of samples. Specifically, the threshold is lowered for classes with fewer samples and raised for classes with more samples. This approach helps to mitigate overfitting in classes with abundant samples and biased predictions in classes with limited samples, keeping the sample number of each class in a balanced state. The implementation of these steps allows us to establish distinct thresholds for each class, thereby enhancing the generalization ability and performance of the model. The overall process is summarized in Algorithm 1.

TABLE I
DETAILED CATEGORIES AND SAMPLES NUMBERS OF THE HOUSTON2013 DATASET

No.	Land-cover	Train	Test
1	Health grass	4	1251
2	Stressed grass	4	1254
3	Synthetic grass	3	697
4	Tree	4	1244
5	Soil	4	1242
6	Water	1	325
7	Residential	4	1268
8	Commercial	4	1244
9	Road	4	1252
10	Highway	4	1227
11	Railway	4	1235
12	Park lot 1	4	1233
13	Park lot 2	2	469
14	Tennis court	2	428
15	Running track	2	660
	total	50	15029

TABLE II
DETAILED CATEGORIES AND SAMPLES NUMBERS OF THE MUUFL DATASET

No.	Land-cover	Train	Test
1	Trees	70	23246
2	Mostly grass	13	4270
3	Mixed ground surface	21	6882
4	Dirt and sand	6	1826
5	Road	21	6687
6	Water	2	466
7	Building shadow	7	2233
8	Building	19	6240
9	Sidewalk	5	1385
10	Yellow curb	1	183
11	Cloth panels	1	269
	total	166	53687

TABLE III
DETAILED CATEGORIES AND SAMPLES NUMBERS OF THE TRENTO DATASET

No.	Land-cover	Train	Test
1	Apple trees	13	4034
2	Buildings	9	2903
3	Ground	2	479
4	Woods	28	9123
5	Vineyard	32	10501
6	Roads	10	3174
	total	94	30214

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (×100) FOR THE HOUSTON2013 DATASET

Class	S2ENet	GLT-Net	HCT-Net	NNC-Net	Sal^{2}RN	FreeMatch	FlexMatch	UncertainGCN	Proposed
Health grass	96.72	84.49	79.38	78.42	94.48	85.77	84.41	85.61	84.89
Stressed grass	85.09	64.51	77.35	82.62	87.16	99.28	97.77	97.05	98.17
Synthetic grass	98.57	96.27	90.39	99.71	94.98	99.14	98.71	99.57	99.28
Tree	91.96	93.25	93.89	77.33	94.29	98.79	99.28	93.09	99.76
Soil	98.95	98.79	97.75	99.11	93.16	100.00	100.00	96.38	100.00
Water	72.92	66.77	85.23	69.23	19.08	79.69	82.77	75.38	82.77
Residential	74.37	66.64	83.36	80.84	85.25	90.69	94.48	73.90	91.09
Commercial	46.86	91.08	40.03	66.88	58.36	68.17	65.11	51.37	72.83
Road	81.63	84.11	63.34	53.91	74.36	85.30	75.56	80.03	88.50
Highway	64.63	66.67	62.51	74.08	61.86	55.58	53.79	57.13	76.77
Railway	92.47	75.06	91.90	87.13	91.58	88.66	87.37	85.43	81.94
Park lot 1	39.90	50.93	72.18	61.64	44.93	62.45	57.10	61.31	52.39
Park lot 2	92.11	71.64	93.60	75.48	82.73	3.20	9.59	69.30	11.09
Tennis court	100.00	90.19	99.77	100.00	94.86	100.00	99.77	100.00	100.00
Running track	100.00	100.00	100.00	99.55	89.24	100.00	100.00	100.00	100.00
OA (%)	80.30	79.36	79.30	78.82	79.14	82.87	81.48	80.41	**84.14**
AA (%)	82.41	80.03	82.05	80.40	77.75	81.12	80.38	81.70	**82.63**
kappa ×100	78.72	77.71	77.62	77.09	77.42	81.46	79.95	78.82	**82.84**

IV. EXPERIMENT RESULTS AND DISCUSSION

To prove the effectiveness of SASS, we perform extensive experiments on Houston2013, Trento, and MUUFL Gulfport datasets. We first present detailed information about three datasets used in the experiment. Then, we compare SASS with other algorithms from both subjective and objective perspectives. Finally, we perform an ablation experiment to prove the effectiveness of different components.

A. Data Description

1) Houston 2013 Dataset: This dataset was provided at the 2013 IEEE GRSS Data Fusion Contest [43]. It covers the University of Houston campus and adjacent urban areas. The HSI and LiDAR data both have 349 × 1905 pixels, with a spatial resolution of 2.5 m. The HSI contains 144 spectral bands ranging from 0.38 to 1.05 μm; the LiDAR contains 1 band. The dataset includes labels of 15 classes and 15 029 samples in total. The detailed categories and the count of samples are shown in Table I.

2) MUUFL Gulfport Dataset: The dataset was taken at the University of Southern Mississippi Gulf Coast campus. Both HSI and LiDAR consist of 325 × 220 pixels, with a spatial resolution of 1 m. The HSI contains 64 spectral bands ranging from 0.38 to 1.05 μm; the LiDAR contains two bands. The dataset includes 11 class labels, with a total of 53 687 samples. The detailed categories and the count of samples are shown in Table II.

3) Trento Dataset: This dataset covers the rural area of Trento, Italy [44]. It includes HSI and LiDAR data. Both are 166 × 600 pixels, with a resolution of 1 m. The HSI includes 63 bands ranging from 0.42 to 0.99 μm; The LiDAR contains one band. The dataset includes a total of 30 214 samples across six class labels. The detailed categories and the count of samples are shown in Table III. For three datasets, we randomly select 0.3% samples from each category to train the deep learning-based methods.

B. Implementation Details

All experimental results are obtained on NVIDIA RTX 3090 GPU. The deep learning framework versions utilized for the code are torch1.13.1 and dgl2.0.0. We employ an Adam optimizer with learning rates set to 0.0001 for the classifier and 0.0005 for GTN. The batch size is 32, and the max epoch is 300. α, β, γ, n, and d for three datasets are set to 0.01, 0.01, 0.01, 0.3, and 300. We employ overall accuracy (OA), average accuracy (AA), and the Kappa coefficient to evaluate the classification performance of different methods objectively.

To evaluate the effectiveness of SASS, five joint classification methods, two SSL methods, and an active learning method were chosen as the comparison methods. Among these comparison methods, the S2ENet [45], NNCNet [46], and Sal^{2}RN [47] are high-precision joint classification methods; the GLT-Net and HCT are transformer-based classification methods; the FreeMatch [48] and FlexMatch [49] are effective semisupervised methods; and UncertainGCN [50] is an active learning framework.

In the same GPU and deep learning framework versions, we retrain all the comparison methods using the same labeled samples, which are randomly selected from the entire set of labeled samples. The parameters of the comparison methods are kept consistent with the settings in the original paper, and the comparison methods are trained to the optimal results. The optimal result of all tests is presented in the tables of classification performance and the classification result maps.

C. Classification Results

Tables IV–VI present the objective classification results of different methods. From the comparative results in the tables, it can be observed that the classification accuracy of semisupervised and active learning methods is generally higher than supervised methods. While supervised methods perform well with sufficient labels, their classification performance

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TABLE V

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (×100) FOR THE MUUFL DATASET

Class	S2ENet	GLT-Net	HCT-Net	NNC-Net	Sal^{2}RN	FreeMatch	FlexMatch	UncertainGCN	Proposed
Trees	94.97	98.09	92.69	94.28	94.87	97.02	96.71	97.95	97.17
Mostly grass	71.45	74.94	77.10	76.11	61.15	78.67	83.19	76.37	84.22
Mixed ground surface	82.26	76.11	82.42	81.78	85.63	84.99	83.28	84.90	82.88
Dirt and sand	93.76	83.73	85.16	83.02	85.10	83.57	88.44	84.72	87.08
Road	88.98	91.88	80.98	90.46	91.00	93.38	95.09	93.70	95.42
Water	43.13	39.06	99.79	97.64	68.24	87.34	93.13	88.63	92.70
Building shadow	76.31	50.11	62.70	41.96	63.99	79.53	80.43	83.07	83.48
Building	94.10	96.03	96.31	85.56	90.10	97.48	95.58	96.68	97.90
Sidewalk	13.14	40.14	11.26	1.23	13.72	46.43	8.16	22.38	45.49
Yellow curb	0.00	3.83	13.11	4.92	10.93	3.28	0.00	2.73	1.64
Cloth panels	36.80	5.58	31.60	87.36	24.16	43.49	39.03	32.71	44.24
OA (%)	86.63	87.14	84.98	84.47	85.38	90.46	89.69	90.14	**91.30**
AA (%)	63.17	59.95	66.65	67.67	62.63	72.29	69.37	69.44	**73.84**
kappa ×100	82.26	82.74	80.27	79.26	80.48	87.37	86.35	86.88	**88.48**

TABLE VI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (×100) FOR THE TRENTO DATASET

Class	S2ENet	GLT-Net	HCT-Net	NNC-Net	Sal^{2}RN	FreeMatch	FlexMatch	UncertainGCN	Proposed
Apple trees	99.63	99.48	99.68	99.95	98.29	98.59	99.50	99.68	99.80
Buildings	95.32	94.56	92.04	94.80	93.56	98.86	98.62	98.97	99.04
Ground	97.70	78.91	86.01	57.41	91.23	95.82	87.68	94.78	94.15
Woods	100.00	100.00	100.00	100.00	99.97	100.00	100.00	100.00	100.00
Vineyard	100.00	100.00	99.94	99.18	99.99	100.00	99.99	100.00	99.99
Roads	93.82	97.45	88.37	93.51	88.91	96.22	98.24	94.55	97.86
OA (%)	98.82	98.81	97.73	97.85	97.84	99.24	99.42	99.20	**99.56**
AA (%)	97.75	95.07	94.34	90.81	95.32	98.25	97.34	98.00	**98.47**
kappa ×100	98.42	98.40	96.97	97.13	97.11	98.98	99.22	98.94	**99.41**

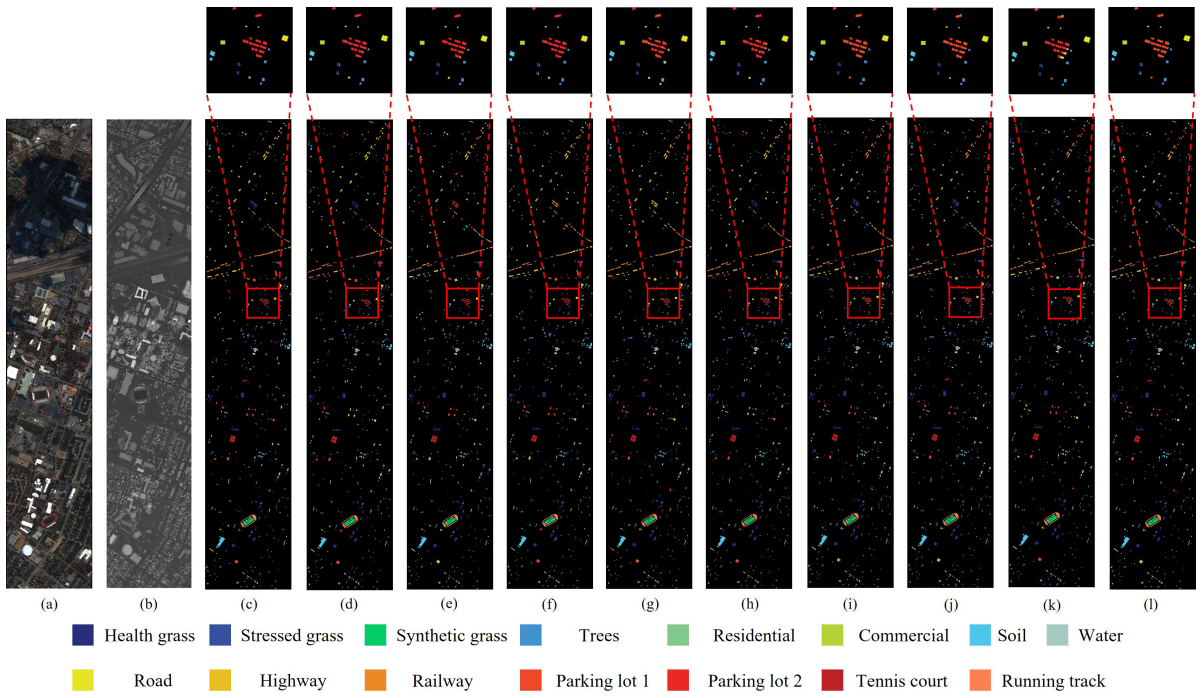


Fig. 4. Classification result maps for different comparison methods on the Houston2013 dataset. (a) Pseudocolor image for HSI. (b) LiDAR DSM map. (c) Ground truth. (d) S2ENet. (e) GLT-Net. (f) HCT-Net. (g) NNCNet. (h) Sal^{2}RN. (i) FreeMatch. (j) FlexMatch. (k) UncertainGCN. (l) Proposed.

decreases when the number of labels is limited. Semisupervised methods have higher accuracy, indicating that effectively utilizing unlabeled data in various ways can significantly improve the classification accuracy when samples are scarce.

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Moreover, our method achieves better results compared to other semisupervised methods. On the Houston2013 dataset, the OA value of SASS is 1.27%, 2.66%, and 3.73% higher than the FreeMatch, FlexMatch, and UncertainGCN, respectively. Similarly, on the MUUFL dataset, SASS achieves an OA of 91.30%, representing improvements of 0.84%, 1.61%, and 1.16% compared to the three methods, respectively. On the Trento dataset, SASS outperforms FreeMatch, FlexMatch, and UncertainGCN with higher OA values of 0.32%, 0.14%, and 0.36%, respectively. Compared to comparison methods, the proposed method shows the highest OA, AA, and Kappa on three datasets. The success can be attributed to the use of SPFD and MSFA, which acquire the shared feature representations across different domains, enhancing the ability of classifiers to fit the distribution of TD. Moreover, the graph transformer network-based class-balanced pseudolabel generation strategy is utilized to select unlabeled samples that are beneficial to improve the classification performance to generate corresponding pseudolabels for model retraining, thus achieving superior performance compared to other approaches. The conclusion is supported by the classification maps shown in Figs. 4–6.

D. Ablation Study

We perform ablation experiments to validate the effectiveness of different components in this section.

1) Effectiveness of SPFD: We design SPFD to capture the shared representations of SD and TD, enabling the model trained on shared features to better adapt to the distribution of TD. To prove the effectiveness of the SPFD, we design a variant of the proposed model (Variant-w/o SPFD), in which only features obtained from labeled data are used to train the model. As shown in Fig. 7, SASS outperforms the Variant-w/o SPFD by 6.57%, 0.75%, and 0.21% in OA across three datasets. In addition, we display the classification result maps and partially enlarged images of Variant-w/o SPFD, SASS, and ground truth in Fig. 8. We can find that the proposed SASS achieves better results than the Variant-w/o SPFD and is closer to the ground truth even in scenes with complex distribution. This indicates that using shared features can improve the ability of the model to fit the distribution of TD.

2) Effectiveness of GTSA Strategy: We design GTSA to query samples beside the decision boundary from the TD, which not only has high-confidence prediction results but also can enhance the generalization of the model. To assess the effectiveness of the GTSA, we design a variant of the proposed model (Variant-w/o GTSA). The GTSA is replaced with the random selection in Variant-w/o GTSA. As shown in Fig. 9, randomly selecting unlabeled samples led to a decrease in OA by 0.67%, 1.65%, and 0.31% compared to SASS, which proves the effectiveness of selecting unlabeled samples that are highly uncertain and easy to identify.

3) Size of Input Patch: We construct $\omega \times \omega$ image patches centered at each pixel as input of the proposed method. The size of input patches significantly influences the classification accuracy. Therefore, we design experiments to discuss the influence of different patch sizes. Fig. 10 shows the effects of different patch sizes (9×9, 11×11, 13×13, 15×15, and 17×17) on classification performance of different datasets.

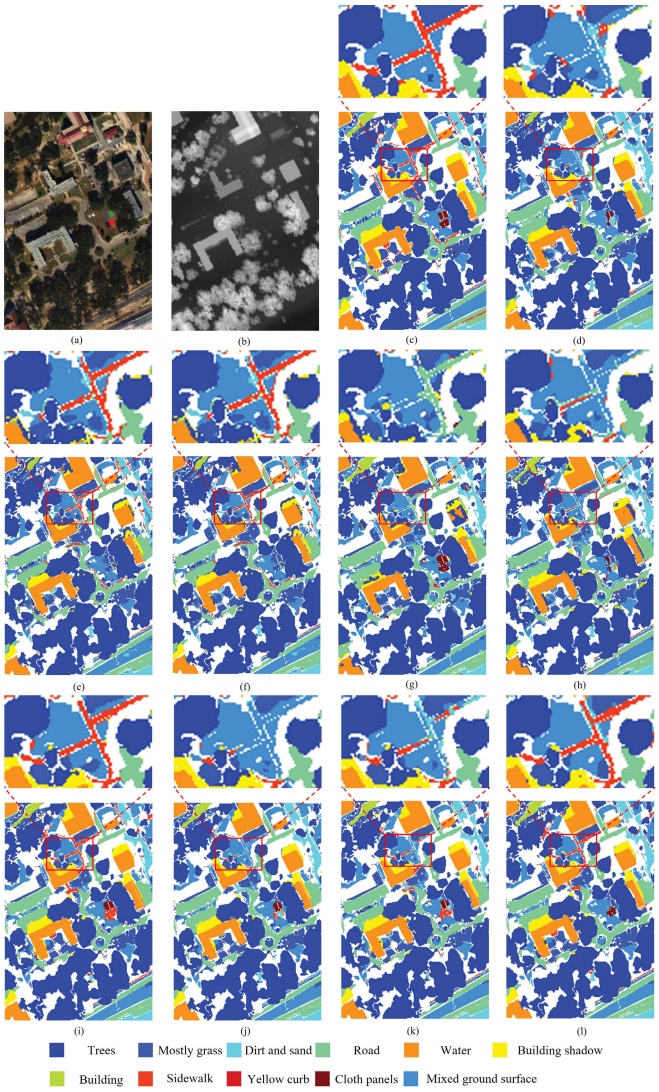


Fig. 5. Classification result maps for different comparison methods on the MUUFL dataset. (a) Pseudocolor image for HSI. (b) LiDAR DSM map. (c) Ground truth. (d) S2ENet. (e) GLT-Net. (f) HCT-Net. (g) NNCNet. (h) Sal^{2}RN. (i) FreeMatch. (j) FlexMatch. (k) UncertainGCN. (l) Proposed.

As shown in Fig. 10, the effects of patch size exhibit similar trends, wherein both excessively larger and small patches lead to a decrease in accuracy. When the input size is too small, the limited amount of information available for the model constrains the classification performance. On the contrary, when the input size is too large, it may introduce pixels with categories different from the central pixel, thereby affecting the feature representation of the central pixel. Therefore, choosing an appropriate size is crucial for the classification performance. According to the experimental results, the optimal input sizes are 13×13.

4) Effects of the Decision Boundary d: This section describes an experiment analyzing the impact of the decision boundary d on the classification performance across three datasets. The experimental results are shown in Fig. 11. The results indicate that both excessively small and large values of d lead to decreased accuracy. A small d value results in high uncertainty of selected samples, leading to lower

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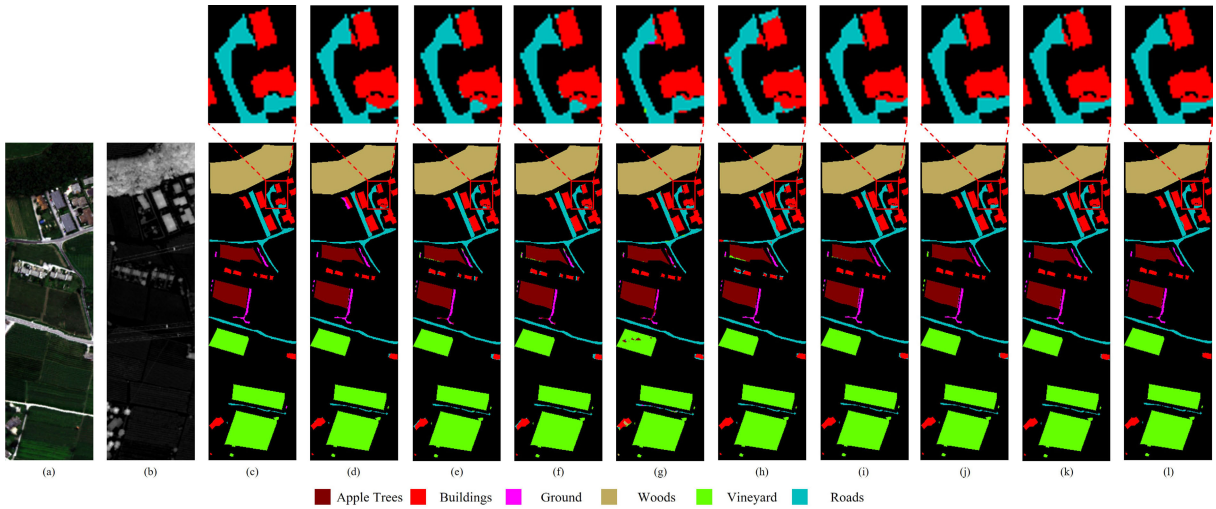


Fig. 6. Classification result maps for different comparison methods on the Houston dataset. (a) Pseudocolor image for HSI. (b) LiDAR DSM map. (c) Ground truth. (d) S2ENet. (e) GLT-Net. (f) HCT-Net. (g) NNCNet. (h) Sal^{2}RN. (i) FreeMatch. (j) FlexMatch. (k) UncertainGCN. (l) Proposed.

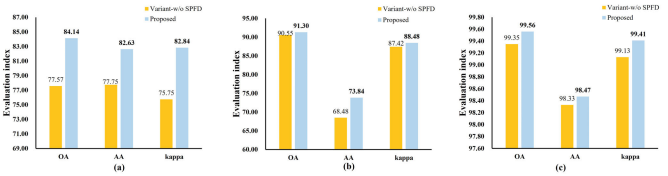


Fig. 7. Effects of SPFD on classification performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

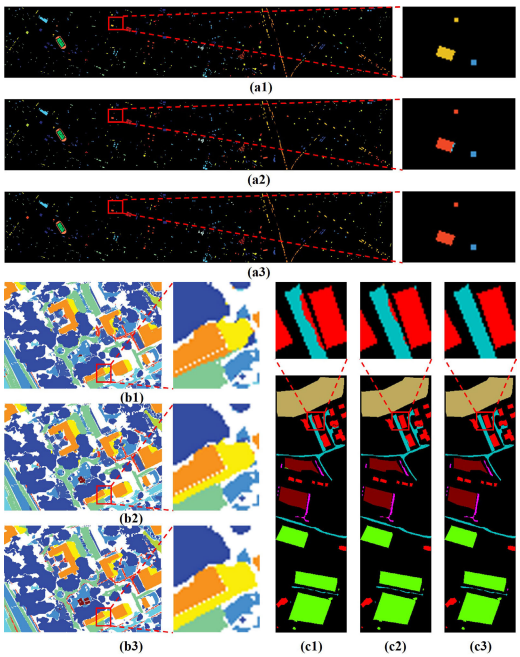


Fig. 8. Classification result maps of different variants. (a1)–(a3) Variant-w/o SPFD, the proposed SASS, and ground truth on the Houston2013 dataset. (b1)–(b3) Variant-w/o SPFD, the proposed SASS, and ground truth on the MUUFL dataset. (c1)–(c3) Variant-w/o SPFD, the proposed SASS, and ground truth on the Trento dataset.

prediction accuracy, while a larger value may cause overfitting by providing pseudolabels similar to labeled samples. Therefore, selecting an appropriate d is crucial. According to the experimental results, the optimal value of d for three datasets is 0.30.

TABLE VII

COMPARISON OF INFERENCE TIME, MODEL PARAMETERS, AND FLOPS ON THE HOUSTON2013 DATASET

	Inference time (s)	Parameters (K)	FLOPS (M)
S2ENet	0.464	270.953	45.772
GLT-Net	1.377	560.771	156.540
HCT-Net	0.438	429.839	7.933
NNCNet	1.104	2812.928	76.136
Sal^{2}RN	0.435	940.797	6.318
proposed	0.550	870.377	133.211

5) Effectiveness of the ATSA Strategy: Because the difficulty of classification varies for different classes, a fixed threshold may favor easier classes, neglecting more challenging ones. To address this problem, we introduce the ATSA to estimate distinct thresholds for different classes. To prove the effectiveness of ATSA, we design a variant of SASS (Variant-w/o ATSA), in which the adaptive class-specific threshold is replaced by a predefined constant threshold. Experimental results across three datasets are depicted in Fig. 12. It is evident that SASS achieves a higher OA by 4.49%, 1.25%, and 0.53%, respectively. The improvement on three objective indicators proves the importance of designing class-specific thresholds according to the category difficulty.

E. Computational Complexity

Table VII illustrates the complexity of five classification models on the Houston2013 dataset, including inference time, the number of trainable weight parameters, and computational cost, which is quantified by floating-point operations per second (FLOPS). It can be observed that the indicators of our method are in the middle level. Despite SASS having a relatively larger number of parameters and FLOPS due to the SPFD and GTSA compared to other methods, it achieves

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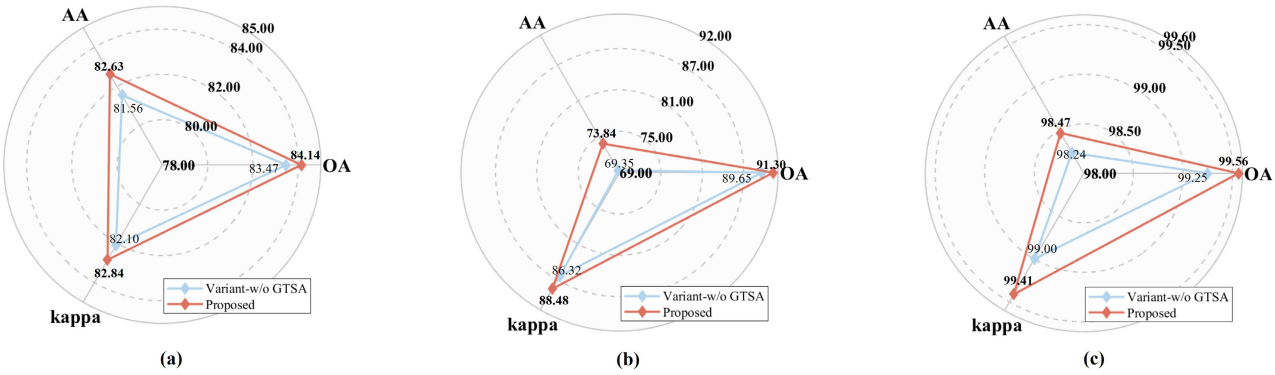


Fig. 9. Effects of GTSA on classification performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

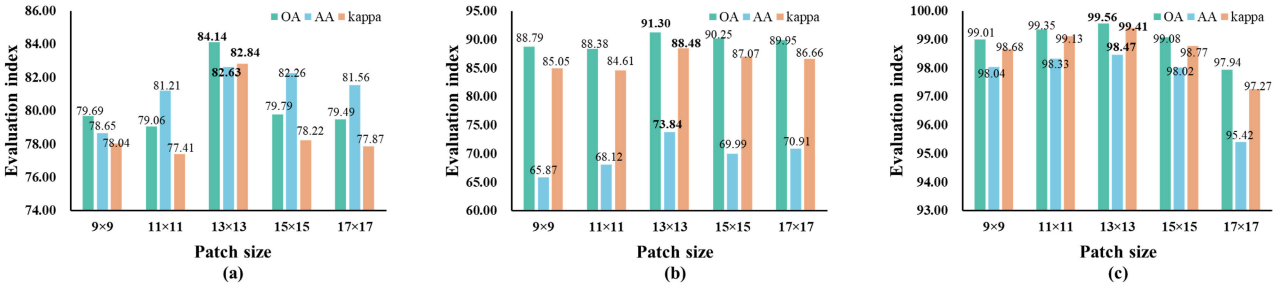


Fig. 10. Effects of the patch size on classification performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

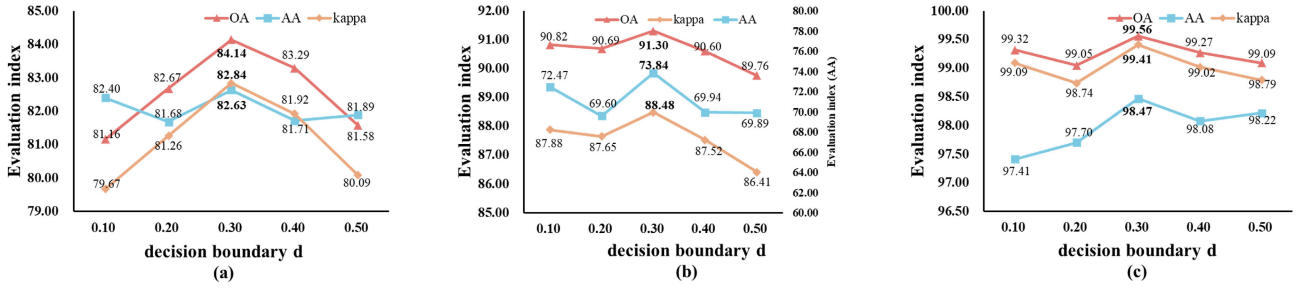


Fig. 11. Effects of the decision boundary d on classification performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

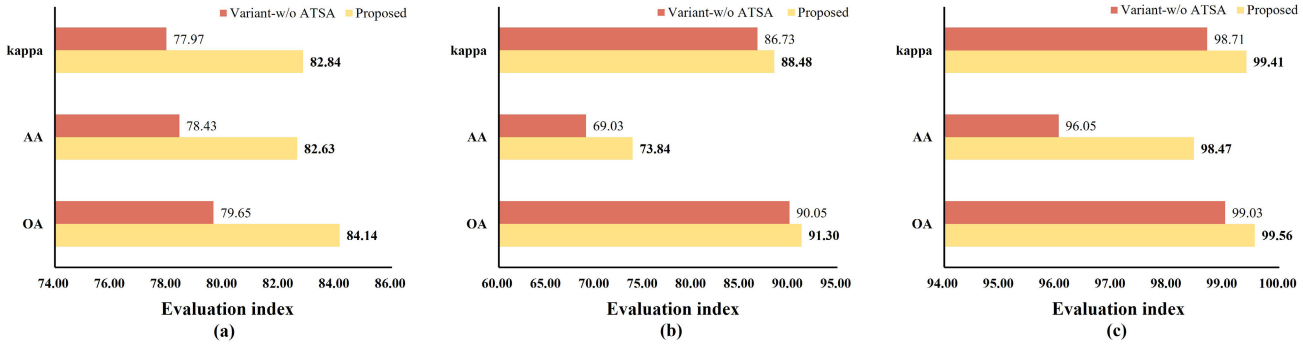


Fig. 12. Effects of ATSA on classification performance. (a) Houston2013 dataset. (b) MUUFL dataset. (c) Trento dataset.

significantly higher accuracy. Overall, the computational cost required by SASS remains within an acceptable range.

V. Conclusion

In this article, we introduce an SASS learning method for HSI and LiDAR classification. We obtain shared representations of SD and TD data via SPFD and MSFA and use these shared features to train the classifier to better model the distribution of TD. Based on these shared features, we employ GTSA to query valuable samples from the TD and generate corresponding pseudolabels using the ATSA. These high-confidence pseudolabels are utilized to retrain the model.

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Experiments on three datasets confirm the superiority of SASS compared to existing methods. In conclusion, our proposed method adeptly leverages the information from unlabeled samples to improve the classification performance of the model.

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